Gabriel MARIADASS

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Summary

M.Sc. candidate in Mathematics, Vision, and Machine Learning (MVA) at ENS Paris-Saclay, combining uncertainty-aware ML research (NTU Singapore, LPSM Paris) with production-oriented ML engineering (Dassault Systèmes, Banque de France). Seeking a CIFRE PhD or an ML/AI role in industry, startup, or scale-up, based in Paris. Available from October 2026.

Experience

LPSM, France & NTU, Singapore *04/2026 – 08/2026*
Research Intern – Possibilistic Variational Inference
Supervisors: Badr-Eddine Chérief-Abdellatif (CNRS, LPSM) and Jérémie Houssineau (NTU)

- Wrote reference notes on uncertainty modeling, variational inference, information theory, and possibility theory
- Reproduced the reference maxitive Donsker–Varadhan (MDV) experiments and derived tractable landscapes on closed-form cases (Bernoulli–Beta, Gaussian, Laplace, bimodal mixtures)
- Designed four anisotropic precision structures for PVI (isotropic, diagonal, low-rank + diagonal, block-diagonal) and analyzed their convergence properties
- Investigated connections between Sharpness-Aware Minimization and PVI

Dassault Systèmes, France *01/2025 – 07/2025*
ML Engineer – Deep Learning & GNN (Molecular Retrosynthesis)

- Migrated GNN retrosynthesis pipeline from DGL to PyTorch Geometric, reducing CPU inference time by 38%
- Improved model accuracy by up to 2 percentage points through Bayesian hyperparameter optimization
- Benchmarked dynamic/static quantization and refactored the preprocessing pipeline for inference optimization

Banque de France, France *09/2024 – 12/2024*
Data Scientist – On-site Credit Institution Control Delegation

- Built neural network credit-scoring models (AUC 0.83) on 700K loan contracts, +0.14 Gini vs. IRB
- Led PD/LGD/EAD risk-parameter and time-series analysis for a live ECB-mandated bank audit

Education

ENS Paris-Saclay, France *2025 – 2026*
M.Sc. Mathematics, Vision, and Machine Learning (MVA), average 17.29/20 (best 8 grades)

- **Key courses:** Deep Learning, Bayesian Machine Learning, Graphs in ML, Advanced Learning for Text and Graph Data, Convex Optimization for ML

École Nationale des Ponts et Chaussées, France *2022 – 2026*
Diplôme d'Ingénieur – Mathematical and Computer Engineering (IMI); final year at ENS Paris-Saclay MVA

Lycée Henri-IV, France *2020 – 2022*
*CPGE – PCSI/PC**

Technical Skills

- **ML:** PyTorch, PyTorch Geometric, Deep Learning, GNNs, Variational Inference, Bayesian Deep Learning, Quantization, Hyperparameter Optimization
- **Mathematics:** Probability, Statistics, Convex Optimization, Stochastic Processes, Possibility Theory
- **Programming & Tools:** Python, SQL, R, SAS, Git, LaTeX

Languages & Interests

- **Languages:** French (native), English (C1), Spanish (B1)
- **Interests:** Fitness, Travel, History, Entrepreneurship